Expected steps

- preliminaries
  - model of the set-up at hand
  - identification of those parameters that are difficult to measure (friction coefficient, spring stiffness, etc.)
  - model validation via time and frequency responses
  - open-loop analysis (stability, observability, controllability, operating range, performance)
- list of self-given specifications for the close-loop system
- for each control task
  - design based on a linearized/nonlinear model
  - test of the controller on the model via simulation
  - controller validation via time and frequency responses
- critical analysis/comparison of all developed control strategies

Control techniques

- Frequency-Based (FB) techniques
  - compute the transfer function $G(s)$ of the linearized system
  - design the $R(s)$ such that $L(s)$ has desired properties
  - design via Bode diagrams (or root locus or Nyquist)
- State-Space (SS) techniques
  - compute the state-space representation of the linearized system
  - design state estimators (Luenberger observer & Kalman filter)
  - design Pole-Placement & Linear-Quadratic controllers
- Advance-Control (AC) techniques, for example:
  - Model Predictive Control (MPC)
  - Sliding Mode
  - Feedback Linearization
  - Adaptive/Robust Control

Time validation

- input impulse/step to $u(t)$ (or $y^\circ(t)$ in close loop)
- record output response $y(t)$ (and $u(t)$ in close loop)
- compare with model impulse/step response
  in terms of gain, settling time, ...

Frequency validation

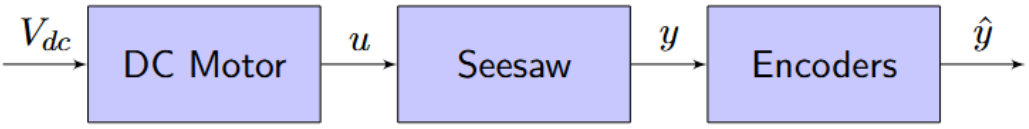
- input sine waves to $u(t)$ (or $y^\circ(t)$ in close loop)
- record output (sine) responses $y(t)$ (and $u(t)$ in close loop)
- compare with model frequency response on a Bode diagram

Expected timeline

Suggestions

- plan activities for the next lab session
- have a backup plan

System



Quantities

- V_{dc} is the voltage applied to the motor
- u is the resulting force applied to the cart
- y contains the cart position and the body angular position
- $\hat{y}$ contains the *relative* cart and body positions

Main challenge

Dynamical model and instability of the horizontal equilibrium position of the body

Control objectives

1. position control for the cart (FB)
2. stabilization of the body in the horizontal position (FB & SS)
3. periodic trajectory tracking for the body and lift-up (SS & AC)

3 IP02 SPECIFICATIONS

Table 3.1 lists and characterizes the main parameters associated with the IP02. Some of these are used in the mathematical model.

Symbol	Description	Value	Variation
V_{nom}	Motor nominal input voltage	6.0 V	
R_m	Motor armature resistance	2.6 Ω	$\pm$ 12%
L_m	Motor armature inductance	0.18 mH	
k_t	Motor current-torque constant	7.68×10^{-3} N m/A	$\pm$ 12%
k_m	Motor back-emf constant	7.68×10^{-3} V/(rad/s)	$\pm$ 12%
η_m	Motor efficiency	0.69	$\pm$ 5%
$J_{m,rotor}$	Rotor moment of inertia	3.90×10^{-7} kg $\cdot$ m^2	$\pm$ 10%
K_g	Planetary gearbox gear ratio	3.71	
η_g	Planetary geabox efficiency	0.90	$\pm$ 10%
M_c	Mass of cart	0.38 kg	
M_w	Mass of cart weight	0.37 kg	
$B_{eq,c}$	Equivalent viscous damping coefficient (Cart)	4.3 N m s/rad	
$B_{eq,c}$	Equivalent viscous damping coefficient (Cart and Weight)	5.4 N m s/rad	
L_t	Track length	0.990 m	
T_c	Cart travel	0.814 m	
P_r	Rack pitch	1.664×10^{-3} m/tooth	
r_{mp}	Motor pinion radius	6.35×10^{-3} m	
N_{mp}	Motor pinion number of teeth	24	
r_{pp}	Position pinion radius	0.01483 m	
N_{pp}	Position pinion number of teeth	56	
K_{ec}	Cart encoder resolution	2.275×10^{-5}	
K_{ep}	Pendulum encoder resolution	0.0015 rad/count	
f_{max}	Maximum input voltage frequency	50 Hz	
I_{max}	Maximum input current	1 A	
ω_{max}	Maximum motor speed	628.3 rad/s	

Table 3.1: IP02 Specifications

参数	数值
电阻 R_m	2.6 Ω
电感 L_m	0.18 mH
力矩常数 k_t	7.68×10^{-3} Nm/A
反电动势常数 k_m	7.68×10^{-3} V/(rad/s)
转子惯量 J_m	3.9×10^{-7} kg·m^2
齿轮比 K_g	3.71
齿轮效率 η_g	0.90
电机齿轮半径 r_{mp}	6.35×10^{-3}m

$$V_{dc} = L_m \dot{i} + R_m i + k_m \omega$$

input V_{dc}

$$\tau_m = k_t i$$

Motor electromagnetic torque τ_m

$$\tau_{out} = k_t i K_g \eta_g$$

Output torque after gear transmission τ_{out}

$$u = \frac{\tau_{out}}{r_{mp}}$$

The cart's linear force obtained by dividing the output torque by the gear radius u

$$u = \frac{k_t K_g \eta_g}{r_{mp}} i$$

Laplace domain

$$u(s) = \frac{k_t K_g \eta_g}{r_{mp}} \frac{1}{L_m s + R_m} (V_{dc}(s) - k_e \omega(s))$$

output u

input u

$$u(s) = \frac{k_t K_g \eta_g}{r_{mp}} \frac{1}{L_m s + R_m} \big(V_{dc}(s) - k_e \omega(s)\big)$$

output

$$y = \begin{bmatrix} x \\ \theta \end{bmatrix}$$

signal flow chain

$$V_{dc} \to u \to x \to \theta \to y$$

参数	数值
小车质量 M_c	0.38 kg
额外砝码质量 M_w	0.37 kg
轨道长度	0.99 m
小车有效行程	0.814 m

参数	数值
小车等效阻尼$B_{eq,c}$	4.3 Nms/rad
小车等效阻尼$B_{eq,c}$ 加砝码	5.4 Nms/rad

Cart Dynamics

$$M_e = M_c + \frac{J_m K_g^2}{r_{mp}^2}$$

$$M_e \ddot{x} + B_c \dot{x} = u$$

Laplace transform

$$(M_e s^2 + B_c s)X(s) = U(s)$$

Substituting $u(s)$ yields

$$(M_e s^2 + B_c s)X(s) = \frac{k_t K_g \eta_g}{r_{mp}} \frac{1}{L_m s + R_m} \big(V_{dc}(s) - k_e \omega(s)\big)$$

The relationship between the motor angular velocity and the cart velocity is

$$\omega = \frac{K_g}{r_{mp}} \dot{x}$$

Laplace domain

$$\omega(s) = \frac{K_g}{r_{mp}} s X(s)$$

substitution

$$(M_e s^2 + B_c s)X(s) = \frac{k_t K_g \eta_g}{r_{mp}} \frac{1}{L_m s + R_m} \left(V_{dc}(s) - k_e \frac{K_g}{r_{mp}} s X(s) \right)$$

rearranging

$$\left[(M_e s^2 + B_c s) + \frac{k_t k_e K_g^2 \eta_g}{r_{mp}^2} \frac{s}{L_m s + R_m} \right] X(s) = \frac{k_t K_g \eta_g}{r_{mp}} \frac{1}{L_m s + R_m} V_{dc}(s)$$

1 position control for the cart (FB)

After rearranging:

$$G_x(s) = \frac{X(s)}{V_{dc}(s)} \frac{\alpha}{(L_m s + R_m)(M_e s^2 + B_c s) + \beta s}$$

Where:

$$\alpha = \frac{k_t K_g \eta_g}{r_{mp}}$$

$$\beta = \frac{k_t k_e K_g^2 \eta_g}{r_{mp}^2}$$

Third-order system

Expand the denominator:

$$(L_m s + R_m)(M_e s^2 + B_c s) + \beta s$$

$$L_m M_e s^3 + (L_m B_c + R_m M_e) s^2 + (R_m B_c + \beta) s$$

Therefore:

$$G_x(s) = \frac{\alpha}{L_m M_e s^3 + (L_m B_c + R_m M_e)s^2 + (R_m B_c + \beta)s}$$

Contains one integrator

Second-order model with the inductance neglected

$$G(s) = \frac{\alpha}{R_m M_e s^2 + (R_m B_c + \beta)s}$$

Where:

$$\alpha = \frac{k_t K_g \eta_g}{r_{mp}}$$

$$\beta = \frac{k_t k_e K_g^2 \eta_g}{r_{mp}^2}$$

Parameter Identification

Factor out s from the denominator:

$$G(s) = \frac{\alpha}{s(R_m M_e s + (R_m B_c + \beta))}$$

The identified model is:

$$G(s) = \frac{\alpha}{R_m B_c + \beta} \cdot \frac{1}{s\left(\frac{R_m M_e}{R_m B_c + \beta} s + 1\right)} = \frac{K}{s(\tau s + 1)}$$

System gain

$$K = \frac{\alpha}{R_m B_c + \beta}$$

Time constant

$$\tau = \frac{R_m M_e}{R_m B_c + \beta}$$

The transfer function parameters were estimated using MATLAB's `tfest` function based on the Prediction Error Method (PEM), which minimizes the squared difference between the measured output and the model output.

Equivalent viscous damping coefficient

$$B_c = \frac{\frac{R_m M_e}{\tau} - \beta}{R_m}$$

Actuator force constant (or voltage-to-force gain)

$$\alpha = K(R_m B_c + \beta)$$

Gearbox efficiency

$$\eta_g = \frac{\alpha r_{mp}}{k_t K_g}$$

The parameters provided in the IP02 datasheet include a variation range due to manufacturing tolerances.
These uncertainties affect the actual system dynamics.

During system identification, the uncertainties caused by parameter variations are implicitly reflected in the identified parameters K and τ, and are subsequently manifested in the equivalent viscous damping coefficient B_c and the gearbox efficiency η_g when the physical parameters are computed.

Controller Design $R(s)$

We need to design:

$$L(s) = R(s)G_x(s)$$

Such that:

- Phase margin $\geq 45^\circ$
- Moderate bandwidth
- Small overshoot
  Controller structure:

$$R(s) = K_p \left(1 + \frac{1}{T_i s} + T_d s\right)$$

Bode Design Procedure (FB)

1. Plot the Bode diagram of $G(s)$
2. Check the phase margin
3. Add a PID controller to adjust phase and steady-state performance
4. Adjust the crossover frequency
   Objective:

- Phase margin around 50°
- Crossover frequency approximately $5 \sim 10\ \mathrm{rad/s}$

The integral term is introduced to eliminate steady-state error, while the derivative term improves the transient response and increases the phase margin.